(AQI) Simple Reflex Agent Documentation

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1. Introduction

This project implements a Simple Reflex Agent that reads environmental parameters from a dataset file and calculates the Air Quality Index (AQI) for the Delhi region. The agent directly maps pollutant concentrations to AQI values using predefined breakpoint rules without learning or storing past states. This is the reason it is called a simple agent as it doesn't store any memory of the previous values.

2. How to Run the Program

Requirements:

- Python 3.x installed

- Dataset file: delhi_air_quality_feature_store_processed.csv

Steps:

1. Place the CSV file and Python script in the same folder.

2. Open Command Prompt in that folder.

3. Run: python delhi_aqi_agent.py and give it at least 20 seconds to run as a lot of data values have to be calculated.

4. The program will display the year (2000-2024), followed by the month (01-12 where 01 is January and 12 is December), and the average AQI value of that month and year.

Example Output:

```
2023 - 02 : Average AQI = 262
2023 - 03 : Average AQI = 262
2023 - 04 : Average AQI = 262
2023 - 05 : Average AQI = 262
2023 - 06 : Average AQI = 262
2023 - 07 : Average AQI = 272
2023 - 08 : Average AQI = 288
2023 - 09 : Average AQI = 296
2023 - 10 : Average AQI = 296
2023 - 11 : Average AQI = 296
2023 - 12 : Average AQI = 296
2024 - 01 : Average AQI = 296
2024 - 02 : Average AQI = 296
2024 - 03 : Average AQI = 296
2024 - 04 : Average AQI = 296
2024 - 05 : Average AQI = 296
2024 - 06 : Average AQI = 296
2024 - 07 : Average AQI = 296
2024 - 08 : Average AQI = 296
2024 - 09 : Average AQI = 296
2024 - 10 : Average AQI = 296
2024 - 11 : Average AQI = 296

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[Done] exited with code=0 in 22.118 seconds
```

3. Pollutants Used in AQI Calculation

- PM2.5

- PM10

- NO_2

- SO_2

- O_3

- CO

4. AQI Calculation Formula

AQI is calculated using linear interpolation: AQI = ((I_high - I_low) / (C_high - C_low)) × (C - C_low) + I_low where C is pollutant concentration, and the I and C values are breakpoint ranges.

5. Breakpoints (EPA Standard)

Each pollutant has predefined concentration ranges mapped to AQI ranges.

Example for PM2.5:

0–12 → AQI 0–50

12.1–35.4 → AQI 51–100

35.5–55.4 → AQI 101–150

55.5–150.4 → AQI 151–200

150.5–250.4 → AQI 201–300

250.5–350.4 → AQI 301–400

350.5–500.4 → AQI 401–500

Similar breakpoint tables are defined for PM10, NO_2, SO_2, O_3, and CO.

6. Final AQI Rule

AQI is calculated separately for each pollutant. The final AQI is the maximum among all calculated AQI values. The pollutant with the highest AQI is called the controlling pollutant.

7. AQI Categories

0–50 → Good

51–100 → Moderate

101–150 → Unhealthy for Sensitive Groups

151–200 → Unhealthy

201–300 → Very Unhealthy

301–500 → Hazardous

8. Simple Reflex Agent Explanation

A Simple Reflex Agent acts only on the current percept. In this project, the percept is the latest pollutant concentration from the dataset. The agent applies condition-action rules (breakpoint

tables) to compute AQI and immediately outputs the result. It does not store past data, learn, or predict future AQI.

9. PEAS Description

Performance Measure: Accurate AQI calculation

Environment: Delhi air quality dataset

Actuators:output displaying AQI

Sensors: CSV file pollutant values

10. Assumptions and Limitations

Assumptions:

- Breakpoints follow EPA standard.

- Missing values are ignored.

Limitations:

- No prediction of future AQI.

- Uses static dataset instead of real-time data.

- No learning mechanism.

11. Agent Architecture

Environment (Dataset File) → Percept (Pollutant Values) → Condition-Action Rules (Breakpoints) → Action (AQI Output)

12. Data Source

Dataset Source:
https://huggingface.co/datasets/abhinavsarkar/delhi_air_quality_feature_store_processed.csv